

Heavy tails: exam-style exercises

Statistics for Insurance, Chapter 3 problems first, full solutions after

Throughout, $\bar{F} = 1 - F$ is the survival function, and “regularly varying with index α ” means $\bar{F}(tx)/\bar{F}(x) \rightarrow t^{-\alpha}$ for every $t > 0$, with $\alpha = 1/\gamma > 0$.

Recipe: how to show a distribution is (or is not) regularly varying

1. **Ratio test.** Compute $\lim_{x \rightarrow \infty} \bar{F}(tx)/\bar{F}(x)$. If the limit is $t^{-\alpha}$, the tail is RV with index α . A limit of 0 means lighter than every power. A limit of 1 means slowly varying ($\alpha = 0$). No limit means not RV.
2. **Factor out the biggest power.** Write $\bar{F}(x) = x^{-\alpha}\ell(x)$ and check that ℓ has a positive limit or is a known slowly varying function (such as $\log x$).
3. **Density only?** Use L'Hôpital, $\bar{F}(tx)/\bar{F}(x) \sim tf(tx)/f(x)$, or von Mises: $xf(x)/\bar{F}(x) \rightarrow \alpha$.
4. **Given as $\exp(\int \dots)$?** Use the representation theorem (Thm 3.7).
5. **To show NOT RV:** find a t where the limit is 0, 1, or does not exist.

Problems

A. Show it is regularly varying and find the index

Problem 1 (log-logistic)

warm-up

Let $\bar{F}(x) = \frac{1}{1 + (x/\sigma)^\kappa}$ for $x \geq 0$, with $\sigma, \kappa > 0$.

- (a) Show that $\bar{F}$ is regularly varying and find α and γ .
- (b) For which $p > 0$ is $\mathbb{E}[Z^p] < \infty$?

Problem 2 (half-Cauchy)

standard

Let $|Z|$ have survival function $\bar{F}(x) = 1 - \frac{2}{\pi} \arctan x$ for $x \geq 0$. Show that $\bar{F}$ is regularly varying with index 1. Does $\mathbb{E}|Z|$ exist? *Hint:* $\arctan x + \arctan(1/x) = \pi/2$ for $x > 0$.

Problem 3 (log-gamma)

standard

Let Z have density $f(x) = \frac{\lambda^k}{\Gamma(k)} (\log x)^{k-1} x^{-\lambda-1}$ for $x > 1$, with $\lambda, k > 0$. (This is $Z = e^Y$ with $Y \sim \text{Gamma}(k, \lambda)$.) Show that Z is regularly varying and find the index.

Problem 4 (mixture)

standard

Let $\bar{F}(x) = px^{-2} + (1-p)x^{-3}$ for $x \geq 1$, with $p \in (0, 1)$.

- (a) Show that $\bar{F}$ is regularly varying and find the index.
- (b) What happens if $p = 0$? State the general principle for mixtures.

Problem 5 (representation theorem)*standard*

Let

$$\bar{F}(z) = \exp\left(-\int_1^z \frac{2 + 1/\log(e+u)}{u} du\right), \quad z \geq 1.$$

- (a) Explain why this is a valid survival function on $[1, \infty)$.
- (b) Show that it is regularly varying and find the index, (i) using Theorem 3.7 and (ii) directly by the ratio test.

B. Show it is NOT regularly varying**Problem 6 (heavy, but all moments finite)***standard*Let $\bar{F}(x) = \exp(-(\log x)^2)$ for $x \geq 1$.

- (a) Show that the distribution is heavy-tailed.
- (b) Show that it is not regularly varying.
- (c) Show that $\mathbb{E}[Z^p] < \infty$ for every $p > 0$. Why can this never happen for an RV distribution?

Problem 7 (a super-heavy tail)*standard*Let $\bar{F}(x) = 1/\log x$ for $x \geq e$ (and $\bar{F}(x) = 1$ for $x < e$).

- (a) Show that $\bar{F}$ is slowly varying, so it is not RV with any index $\alpha > 0$.
- (b) Show that the distribution is long-tailed.
- (c) Show that $\mathbb{E}[Z^p] = \infty$ for every $p > 0$.

Problem 8 (sandwiched between powers, yet not RV)*harder*Let $\bar{F}(x) = \frac{1}{3}x^{-2}(2 + \sin(\log x))$ for $x \geq 1$ and $\bar{F}(x) = 1$ for $x < 1$.

- (a) Show that $\bar{F}$ is decreasing on $[1, \infty)$.
- (b) Show that $\frac{1}{3}x^{-2} \leq \bar{F}(x) \leq x^{-2}$, and deduce that $\mathbb{E}[Z^p] < \infty$ iff $p < 2$.
- (c) Show that $\bar{F}$ is *not* regularly varying.

Problem 9 (classification)*warm-up*

For each distribution, say whether it is heavy-tailed (HT), long-tailed (L) and regularly varying (RV). Give a one-line reason for each answer.

- (i) Exp(1) (ii) Uniform(0, 1) (iii) Pareto, $\bar{F}(x) = x^{-3}$, $x \geq 1$ (iv) Weibull, $\bar{F}(x) = e^{-\sqrt{x}}$

C. Properties of regularly varying distributions**Problem 10 (transformations)***standard*Let $Z \geq 0$ be regularly varying with index α .

- (a) Show that cZ ($c > 0$) is RV with index α .
- (b) Show that Z^p ($p > 0$) is RV with index α/p .
- (c) Let W be independent of Z and RV with index $\beta > \alpha$. Show that $\max(Z, W)$ is RV with index α .
- (d) Show that $\log Z$ is *not* heavy-tailed.

Problem 11 (moments via Potter)*exam classic*

Let $Z \geq 0$ be RV with index α . Using Potter's bounds (Thm 3.5), show that $\mathbb{E}[Z^p] < \infty$ for $p < \alpha$ and $\mathbb{E}[Z^p] = \infty$ for $p > \alpha$. Conclude that Z is heavy-tailed.

D. Estimation**Problem 12 (Pareto MLE = Hill)***exam classic*

Let $Z_1, \dots, Z_m$ be iid with the exact Pareto density $f(z) = \alpha u^\alpha z^{-\alpha-1}$, $z > u$, where u is known.

- (a) Find the MLE of α , and hence of $\gamma = 1/\alpha$.
- (b) Show that $\log(Z_i/u) \sim \text{Exp}(\text{rate } \alpha)$. Find the mean and variance of $\hat{\gamma}$.
- (c) Explain how this motivates the Hill estimator and its limit $N(0, \gamma^2)$.

Problem 13 (Hill by hand)*calculation*

Twelve claims (in €10,000) are, in sorted order:

1.1, 1.3, 1.6, 2.0, 2.4, 2.9, 3.5, 4.4, 6.0, 9.0, 15.0, 40.0.

- (a) Compute the Hill estimate $\gamma_{12,5}$ and $\hat{\alpha} = 1/\gamma_{12,5}$.
- (b) Give an approximate 95% confidence interval for γ . Comment.
- (c) Estimate the 99% quantile, and $\mathbb{P}(Z > 100)$, using the Pareto tail approximation.

Solutions

Solution 1 (log-logistic)

(a) *Ratio test.* Divide numerator and denominator by $(x/\sigma)^\kappa$:

$$\frac{\overline{F}(tx)}{\overline{F}(x)} = \frac{1 + (x/\sigma)^\kappa}{1 + (tx/\sigma)^\kappa} = \frac{(\sigma/x)^\kappa + 1}{(\sigma/x)^\kappa + t^\kappa} \rightarrow \frac{1}{t^\kappa} = t^{-\kappa}.$$

So $\overline{F}$ is RV with $\alpha = \kappa$ and $\gamma = 1/\kappa$. (This is the Burr distribution with $\beta = 1$.)

(b) RV with index κ gives $\mathbb{E}Z^p < \infty$ iff $p < \kappa$ (Problem 11). Directly: $\mathbb{E}Z^p = \int_0^\infty px^{p-1}\overline{F}(x)dx$, and for large x the integrand behaves like $p\sigma^\kappa x^{p-1-\kappa}$. This is integrable at ∞ iff $p - 1 - \kappa < -1$, i.e. $p < \kappa$.

Solution 2 (half-Cauchy)

By the hint, $\overline{F}(x) = \frac{2}{\pi}(\frac{\pi}{2} - \arctan x) = \frac{2}{\pi} \arctan(1/x)$. Since $\arctan y \sim y$ as $y \rightarrow 0$,

$$\overline{F}(x) = x^{-1} \cdot \underbrace{\frac{2}{\pi} x \arctan(1/x)}_{\ell(x) \rightarrow 2/\pi}.$$

ℓ has a positive limit, so it is slowly varying. Hence $\overline{F}$ is RV with index 1 ($\gamma = 1$).

Check with von Mises: $f(x) = \frac{2}{\pi(1+x^2)}$, so $\frac{xf(x)}{\overline{F}(x)} \approx \frac{2/(\pi x)}{2/(\pi x)} \rightarrow 1$.

Mean: the index is 1, so we are at the boundary $p = \alpha$. Directly, $\mathbb{E}|Z| = \int_0^\infty \overline{F}(x)dx$, and $\overline{F}(x) \sim \frac{2}{\pi x}$, whose integral diverges. So $\mathbb{E}|Z| = \infty$.

Solution 3 (log-gamma)

We only have the density, so use L'Hôpital. Both $\overline{F}(tx)$ and $\overline{F}(x)$ tend to 0, and $\frac{d}{dx}\overline{F}(tx) = -tf(tx)$:

$$\lim \frac{\overline{F}(tx)}{\overline{F}(x)} = \lim \frac{tf(tx)}{f(x)} = t \cdot \left(\frac{\log(tx)}{\log x} \right)^{k-1} \cdot \frac{(tx)^{-\lambda-1}}{x^{-\lambda-1}} = t^{-\lambda} \cdot \underbrace{\left(1 + \frac{\log t}{\log x} \right)^{k-1}}_{\rightarrow 1} \rightarrow t^{-\lambda}.$$

So Z is RV with index $\alpha = \lambda$, i.e. $\gamma = 1/\lambda$. The shape k only changes the slowly varying part $(\log x)^{k-1}$.

Alternative: $f(x) = x^{-\lambda-1}\ell(x)$ with $\ell(x) = c(\log x)^{k-1}$ slowly varying. Karamata (Thm 3.8) gives $\overline{F}(x) \sim x^{-\lambda}\ell(x)/\lambda$.

Solution 4 (mixture)

(a) Factor out the biggest power, x^{-2} :

$$\overline{F}(x) = x^{-2}(p + (1-p)x^{-1}), \quad \ell(x) = p + (1-p)x^{-1} \rightarrow p > 0.$$

So the index is 2.

(b) If $p = 0$, then $\overline{F}(x) = x^{-3}$ and the index is 3. *Principle:* in a mixture, the **heaviest component decides the tail**, however small its weight $p > 0$. A 1% share of very heavy claims determines the tail index of the whole portfolio.

Solution 5 (representation theorem)

(a) $\bar{F}(1) = e^0 = 1$. The integrand is positive, so the exponent decreases and $\bar{F}$ decreases. Since $\int_1^z 2/u \, du = 2 \log z \rightarrow \infty$, we get $\bar{F}(z) \rightarrow 0$. It is continuous, so it is a valid survival function.

(b)(i) This is exactly the form of Thm 3.7 with $z_0 = 1$, $c(z) \equiv 1 \rightarrow 1$ and $\delta(u) = -2 - \frac{1}{\log(e+u)} \rightarrow -2$. Hence the tail is RV with index 2.

(b)(ii) Directly:

$$\frac{\bar{F}(tx)}{\bar{F}(x)} = \exp\left(-\int_x^{tx} \frac{2}{u} du\right) \exp\left(-\int_x^{tx} \frac{du}{u \log(e+u)}\right) = t^{-2} \cdot \exp(-R_x).$$

For $t > 1$: $0 \leq R_x \leq \frac{1}{\log(e+x)} \int_x^{tx} \frac{du}{u} = \frac{\log t}{\log(e+x)} \rightarrow 0$. So the ratio tends to t^{-2} . The case $t < 1$ is the same.

Solution 6 (heavy, but all moments finite)

(a) For every $t > 0$: $e^{tx} \bar{F}(x) = \exp(tx - (\log x)^2) \rightarrow \infty$, since $(\log x)^2$ grows much more slowly than tx . By (3.1) the distribution is heavy-tailed.

(b) Write $L = \log x$. For $t > 1$:

$$\frac{\bar{F}(tx)}{\bar{F}(x)} = \exp(-(L + \log t)^2 + L^2) = \exp(-2L \log t - (\log t)^2) \rightarrow 0.$$

The limit is 0, not $t^{-\alpha}$, so the tail is not RV.

(c) Substitute $y = \log x$ (so $x = e^y$, $dx = e^y dy$):

$$\mathbb{E}Z^p = 1 + \int_1^\infty p x^{p-1} e^{-(\log x)^2} dx = 1 + \int_0^\infty p e^{py-y^2} dy < \infty,$$

because $-y^2$ beats py . For RV this is impossible: Problem 11 gives $\mathbb{E}Z^p = \infty$ for $p > \alpha$.
Lesson: heavy-tailed does **not** imply infinite moments. The lognormal behaves like this too.

Solution 7 (a super-heavy tail)

(a) $\frac{\bar{F}(tx)}{\bar{F}(x)} = \frac{\log x}{\log x + \log t} \rightarrow 1$ for every $t > 0$. So $\bar{F}$ is slowly varying ($\alpha = 0$), not $t^{-\alpha}$ with $\alpha > 0$. It is heavier than every Pareto tail.

(b) $\frac{\bar{F}(x+y)}{\bar{F}(x)} = \frac{\log x}{\log(x+y)} \rightarrow 1$, since $\log(x+y) - \log x = \log(1+y/x) \rightarrow 0$.

(c) $\mathbb{E}Z^p \geq \int_e^\infty p x^{p-1} \frac{1}{\log x} dx = \infty$, since $\frac{x^{p-1}}{\log x} = x^{-1} \cdot \frac{x^p}{\log x} \geq x^{-1}$ once $x^p \geq \log x$, and $\int_e^\infty x^{-1} dx = \infty$. Even $\mathbb{E}[\log Z] = \infty$. The Hill estimator would diverge here, because “ $\gamma = \infty$ ”.

Solution 8 (sandwiched between powers, yet not RV)

(a) For $x > 1$:

$$\frac{d}{dx} \left[x^{-2} (2 + \sin \log x) \right] = x^{-3} \left[-2(2 + \sin \log x) + \cos \log x \right] \leq x^{-3} [-2 \cdot 1 + 1] < 0.$$

(b) $1 \leq 2 + \sin \log x \leq 3$ gives $\frac{1}{3}x^{-2} \leq \bar{F}(x) \leq x^{-2}$. Then $\mathbb{E}Z^p = \int p x^{p-1} \bar{F}$ is squeezed between two constant multiples of $\int^\infty x^{p-3} dx$. This is finite iff $p < 2$.

(c) $\frac{\bar{F}(tx)}{\bar{F}(x)} = t^{-2} \frac{2 + \sin(\log x + \log t)}{2 + \sin \log x}$. Take $t = e^{\pi/2}$.

- Along $x_n = e^{2\pi n}$: $\sin \log x_n = 0$, so the ratio is $t^{-2} \frac{2+1}{2} = \frac{3}{2}t^{-2}$.
- Along $x_n = e^{2\pi n + \pi/2}$: $\sin = 1$, so the ratio is $t^{-2} \frac{2+0}{3} = \frac{2}{3}t^{-2}$.

The two subsequences give different limits, so the ratio has no limit and $\bar{F}$ is not RV. *Lesson:* being between two powers (Potter-type bounds) is **not** enough for RV. You need an actual limit.

Solution 9 (classification)

	HT	L	RV	reason
Exp(1)	no	no	no	$M(t) = 1/(1-t) < \infty$ for $t < 1$; $\bar{F}(x+1)/\bar{F}(x) = e^{-1} \neq 1$.
Uniform(0,1)	no	no	no	bounded, so $M(t) < \infty$ for every t ; $\bar{F} = 0$ beyond 1.
Pareto x^{-3}	yes	yes	yes	$\bar{F}(tx)/\bar{F}(x) = t^{-3}$ exactly, and $\text{RV} \subset \text{S} \subset \text{L} \subset \text{HT}$.
Weibull $e^{-\sqrt{x}}$	yes	yes	no	$a = \frac{1}{2} < 1$ (Exercise 4); $\sqrt{x+1} - \sqrt{x} \rightarrow 0$; ratio $\rightarrow 0$.

Solution 10 (transformations)

Let $\bar{F}$ be the tail of Z .

(a) $\frac{\mathbb{P}(cZ > tx)}{\mathbb{P}(cZ > x)} = \frac{\bar{F}(t \cdot x/c)}{\bar{F}(x/c)} \rightarrow t^{-\alpha}$, because $x/c \rightarrow \infty$.

(b) $\mathbb{P}(Z^p > x) = \bar{F}(x^{1/p})$, so $\frac{\bar{F}(t^{1/p}x^{1/p})}{\bar{F}(x^{1/p})} \rightarrow (t^{1/p})^{-\alpha} = t^{-\alpha/p}$. The index is α/p . (For example, squaring halves the tail index.)

(c) $\mathbb{P}(\max > x) = 1 - (1 - \bar{F}_Z)(1 - \bar{F}_W) = \bar{F}_Z(x) + \bar{F}_W(x) - \bar{F}_Z(x)\bar{F}_W(x)$. Divide by $\bar{F}_Z(x)$:

$$\frac{\mathbb{P}(\max > x)}{\bar{F}_Z(x)} = 1 + \frac{\bar{F}_W(x)}{\bar{F}_Z(x)}(1 - \bar{F}_Z(x)), \quad \frac{\bar{F}_W(x)}{\bar{F}_Z(x)} = x^{\alpha-\beta} \frac{\ell_W(x)}{\ell_Z(x)} \rightarrow 0,$$

since ℓ_W/ℓ_Z is slowly varying and a negative power beats it. So $\mathbb{P}(\max > x) \sim \bar{F}_Z(x)$, which is RV with index α . Again the heavier tail wins.

(d) Heavy-tailedness is about the right tail, so look at $Y = (\log Z)_+ = \max(\log Z, 0)$. For $0 < t < \alpha$,

$$\mathbb{E}[e^{tY}] \leq 1 + \mathbb{E}[Z^t] < \infty \quad (\text{Problem 11}).$$

For $t \leq 0$, $e^{tY} \leq 1$. So the mgf is finite on a neighbourhood of 0, and $\log Z$ is light-tailed. In fact $\mathbb{P}(\log Z > y) = e^{-\alpha y} \ell(e^y)$, which is roughly an exponential tail. This is why Hill works on logarithms.

Solution 11 (moments via Potter)

Use $\mathbb{E}Z^p = \int_0^\infty px^{p-1}\bar{F}(x) dx$.

$p < \alpha$: take $\varepsilon = (\alpha - p)/2 \in (0, \alpha)$. Potter gives $\bar{F}(x) \leq Cx^{-\alpha+\varepsilon}$, so

$$\mathbb{E}Z^p \leq \int_0^1 px^{p-1} dx + pC \int_1^\infty x^{p-1-\alpha+\varepsilon} dx = 1 + pC \int_1^\infty x^{-1-\frac{\alpha-p}{2}} dx < \infty.$$

$p > \alpha$: take $\varepsilon = (p - \alpha)/2$ (shrink it if needed so that $\varepsilon < \alpha$; that only helps). Potter gives $\bar{F}(x) \geq c x^{-\alpha-\varepsilon}$ for $x \geq 1$, so

$$\mathbb{E}Z^p \geq pc \int_1^\infty x^{p-1-\alpha-\varepsilon} dx = pc \int_1^\infty x^{-1+\frac{p-\alpha}{2}} dx = \infty.$$

Heavy tail: $e^{tx}\bar{F}(x) \geq c e^{tx} x^{-\alpha-\varepsilon} \rightarrow \infty$ for every $t > 0$. Alternatively, $\mathbb{E}e^{tZ} \geq \mathbb{E}[(tZ)^n/n!] = \infty$ for any integer $n > \alpha$.

Solution 12 (Pareto MLE = Hill)

(a) The log-likelihood is $\log L(\alpha) = m \log \alpha + m\alpha \log u - (\alpha + 1) \sum_i \log Z_i$. Setting the derivative to zero:

$$\frac{m}{\alpha} + m \log u - \sum \log Z_i = 0 \implies \frac{1}{\hat{\alpha}} = \frac{1}{m} \sum_{i=1}^m \log \frac{Z_i}{u} =: \hat{\gamma}.$$

The second derivative is $-m/\alpha^2 < 0$, so this is a maximum. By invariance of the MLE, $\hat{\gamma} = 1/\hat{\alpha}$.

(b) $\mathbb{P}(\log(Z/u) > y) = \mathbb{P}(Z > ue^y) = (ue^y/u)^{-\alpha} = e^{-\alpha y}$, which is $\text{Exp}(\text{rate } \alpha)$ with mean γ and variance γ^2 . Hence $\mathbb{E}\hat{\gamma} = \gamma$ (unbiased) and $\text{Var } \hat{\gamma} = \gamma^2/m$.

(c) An RV tail is approximately Pareto above a high threshold. Hill uses the top k observations as “the $m = k$ exceedances” and the random threshold $u = Z_{n,n-k}$. The formula is identical to (a), and the CLT for (b) gives $\sqrt{k}(\hat{\gamma} - \gamma) \approx N(0, \gamma^2)$, which matches Thm 3.14.

Solution 13 (Hill by hand)

(a) $n = 12, k = 5$. The threshold is the $(k + 1)$ -th largest value, $Z_{12,7} = 3.5$. The top five are 40, 15, 9, 6, 4.4.

Z	ratio to 3.5	log ratio
40	11.429	2.4361
15	4.286	1.4553
9	2.571	0.9445
6	1.714	0.5390
4.4	1.257	0.2288
sum		5.6037

$\gamma_{12,5} = 5.6037/5 = \mathbf{1.1207}$ and $\hat{\alpha} = 1/1.1207 = \mathbf{0.892}$. With $\alpha < 1$, the mean claim appears to be *infinite*.

(b) From $\sqrt{k}(\hat{\gamma} - \gamma) \approx N(0, \gamma^2)$, the standard error is $\hat{\gamma}/\sqrt{k} = 1.1207/\sqrt{5} = 0.501$:

$$1.1207 \pm 1.96 \cdot 0.501 = [0.14, 2.10].$$

The interval is very wide because $k = 5$ is tiny. In practice, use a Hill plot and a larger k in its stable region.

(c) Above the threshold, $\bar{F}(x) \approx \bar{F}(u) (x/u)^{-1/\gamma}$ with $\bar{F}(u) \approx k/n$. This gives

$$\hat{\mathbb{P}}(Z > x) = \frac{k}{n} \left(\frac{x}{Z_{n,n-k}} \right)^{-1/\hat{\gamma}}, \quad \hat{q}_p = Z_{n,n-k} \left(\frac{k}{n(1-p)} \right)^{\hat{\gamma}} \quad (\text{set } \hat{\mathbb{P}} = 1 - p \text{ and solve}).$$

- $\hat{q}_{0.99} = 3.5 \cdot \left(\frac{5}{12 \cdot 0.01} \right)^{1.1207} = 3.5 \cdot 41.67^{1.1207} \approx 3.5 \cdot 65.4 \approx \mathbf{229}$ (i.e. €2.29m).
- $\hat{\mathbb{P}}(Z > 100) = \frac{5}{12} \cdot (100/3.5)^{-0.892} = 0.4167 \cdot 28.57^{-0.892} \approx 0.4167 \cdot 0.0502 \approx \mathbf{0.021}$.

Note that the largest observation is 40. The empirical distribution would say $\mathbb{P}(Z > 100) = 0$; the Pareto tail lets us extrapolate beyond the data.

Calculator tip: $a^b = e^{b \ln a}$. For example, $41.67^{1.1207} = e^{1.1207 \cdot 3.7297} = e^{4.180} \approx 65.4$.